DAWOOD UNIVERSITY OF ENGINEERING & TECHNOLOGY

Karachi, Pakistan

P R O J E C T   R E P O R T

Credit Card Fraud Detection

Using Exploratory Data Analysis & Machine Learning

Subject	Introduction to Data Science (ITDS)
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University	Dawood University of Engineering & Technology, Karachi
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Abstract

Credit card fraud is a major issue in the digital financial world, causing significant losses annually. This project develops a machine learning-based fraud detection system using a real-world dataset containing 339,607 transactions. The pipeline includes data preprocessing, exploratory data analysis (EDA), feature engineering, and handling class imbalance through undersampling. A Random Forest Classifier was trained and evaluated achieving 87.50% accuracy and 88.09% cross-validation accuracy. Results demonstrate that combining feature engineering with machine learning significantly improves fraud detection capability.

1. Introduction

This project, developed for the Introduction to Data Science course at Dawood University of Engineering and Technology, Karachi, builds a credit card fraud detection system. The pipeline covers data exploration, preprocessing, feature engineering, and machine learning to classify transactions as fraudulent or legitimate.

Objectives

- Explore and understand patterns in the credit card transaction dataset
- Apply data preprocessing, class balancing, and feature engineering
- Build and evaluate a Random Forest model for fraud classification
- Derive actionable insights through 10 Exploratory Data Analysis (EDA) questions

2. Dataset Description

The dataset (credit_card_frauds.csv) contains labeled real-world credit card transaction records.

Attribute	Details
Total Records	339,607 rows → 4,000 rows after undersampling
Total Columns	15 columns (original) → 8 features for modelling
Data Types	Float64, Int64, Object (String)
Missing Values	0 — Dataset is completely clean
Class Imbalance	Only 0.52% fraud (1,782 of 339,607 rows)
Target Variable	is_fraud (0 = Legitimate, 1 = Fraud)

2.1 Column Definitions

Column	Type	Description
trans_date_trans_time	DateTime	Transaction timestamp — hour, day, month, year extracted
merchant	String	Merchant name — dropped (high cardinality, ~693 unique)
category	String	Merchant category — one-hot encoded (14 unique values)
amt	Float	Transaction amount in USD — strongest predictor (r = 0.64)
city / state	String	Billing location — used in EDA; state one-hot encoded
lat / long	Float	Customer coordinates — used to compute distance feature

city_pop	Int	City population — used in EDA analysis
job	String	Occupation — dropped (weak predictor, high cardinality)
dob	DateTime	Date of birth — parsed to engineer customer age feature
trans_num	String	Unique transaction ID — dropped (non-predictive)
merch_lat / merch_long	Float	Merchant coordinates — used in distance computation
is_fraud	Int	Target: 0 = Legitimate, 1 = Fraudulent

3. Data Preprocessing

- Dataset loaded and inspected using df.info(), df.shape, df.head(), and df.describe().
- Confirmed zero null values across all 15 columns — no imputation required.
- Converted trans_date_trans_time and dob from string to datetime format using pd.to_datetime().
- Dropped trans_num (unique transaction ID), merchant (high cardinality), and job (weak predictor).

4. Handling Class Imbalance

The original dataset contained 337,825 legitimate and only 1,782 fraud transactions (0.52% fraud rate). Such severe imbalance causes models to be biased toward predicting non-fraud. Undersampling was applied: all 1,782 fraud rows were retained, and 2,218 non-fraud rows were randomly sampled (random_state=42), yielding a balanced 4,000-row dataset.

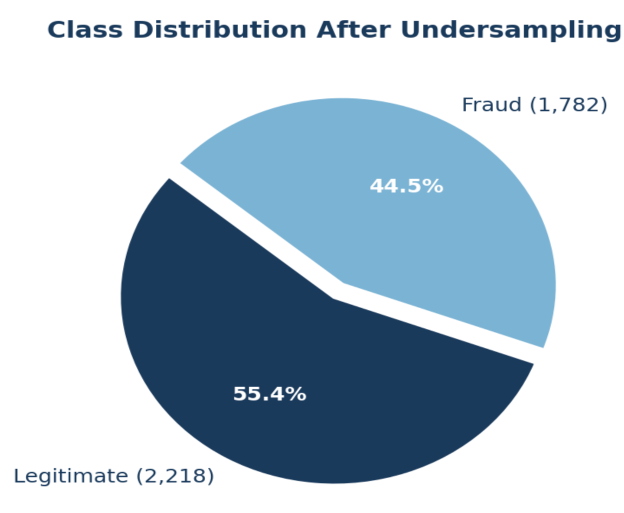

Figure 1 — Class distribution after undersampling (1,782 Fraud vs 2,218 Legitimate)

5. Feature Engineering

Four new features were derived from existing columns to enhance model predictive power:

Feature	Derivation	Reason
age	2026 − birth year from dob	Fraud propensity varies by customer age group
distance	Euclidean distance between customer and merchant coordinates	Fraud often occurs far from customer's usual location
hours	Hour extracted from trans_date_trans_time	Fraud peaks sharply at late-night hours (22–23)
day	Day-of-week from trans_date_trans_time	Weekly patterns affect fraud likelihood

After extraction, source columns (lat, long, dob, merch_lat, merch_long, trans_date_trans_time) were dropped, leaving 8 final features for modelling.

6. Exploratory Data Analysis (EDA)

Ten analytical questions were investigated to uncover fraud patterns in the data.

Q1 — What does the transaction amount distribution indicate?

The histogram shows a strong right-skewed distribution. The majority of transactions fall in the 0–200 USD range. Very few high-value transactions exist, but these outliers are critical — transaction amount shows the highest Pearson correlation of 0.64 with the fraud label.

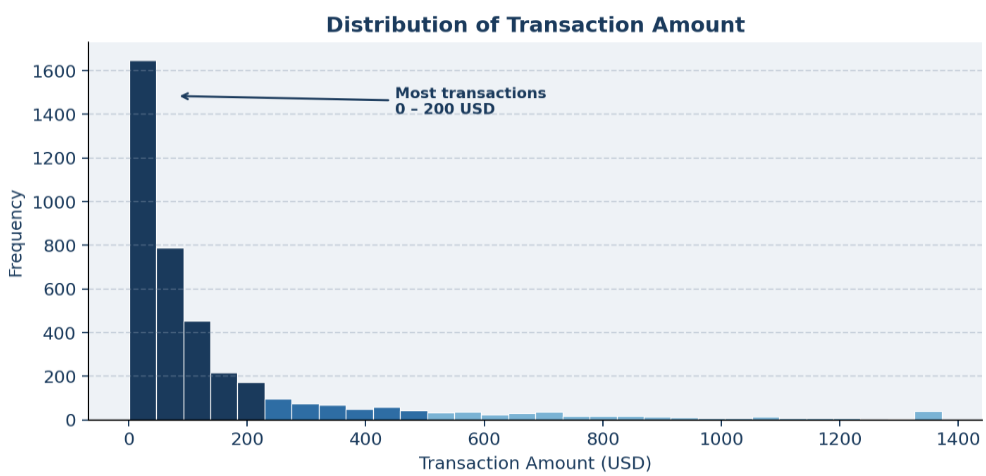

Figure 2 — Distribution of transaction amounts (right-skewed, most in 0–200 USD range)

Q2 — What does the box plot reveal about transaction amounts?

The box plot confirms that most transactions cluster in the lower range with a low median. Numerous high-value outliers are visible, representing unusual or potentially fraudulent transactions that warrant closer investigation.

Q3 — What does the category-wise fraud analysis indicate?

shopping_net has the highest absolute fraud count, followed by grocery_pos. Online shopping and grocery-related transactions are most frequently targeted, suggesting reduced verification in these categories. All 14 categories are represented in the dataset.

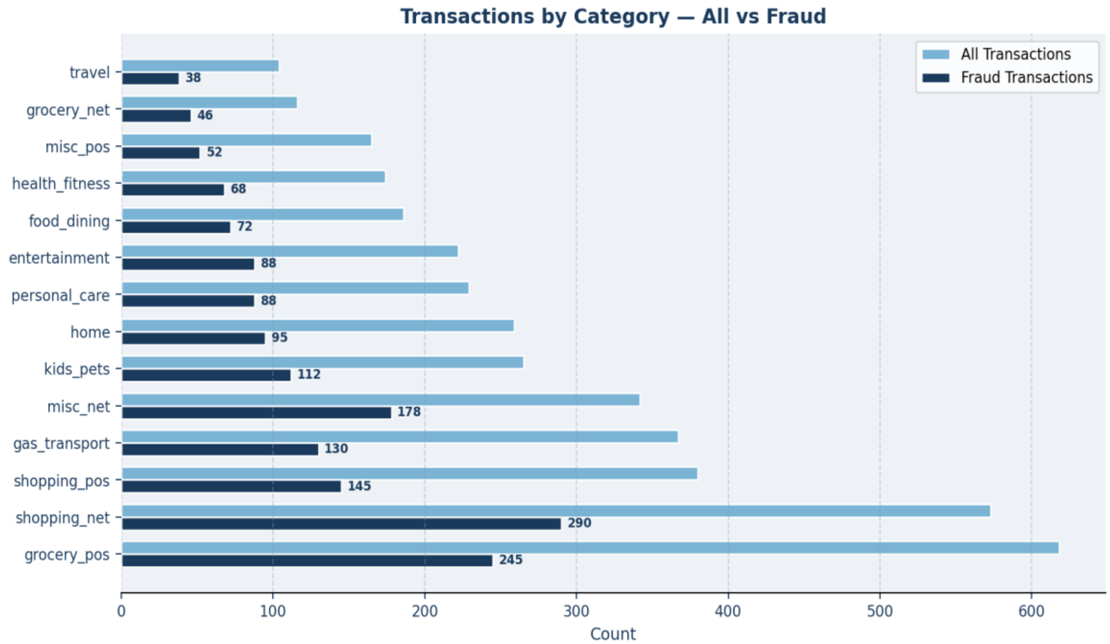

Figure 3 — All transactions vs. fraud transactions by merchant category

Q4 — What does the city-wise fraud analysis show?

176 unique cities appear in the dataset. Albuquerque leads with 24 fraud cases, followed by Aurora (23) and Fort Washakie (21). Fraud incidents are concentrated in certain cities rather than evenly distributed, suggesting location is a useful fraud signal.

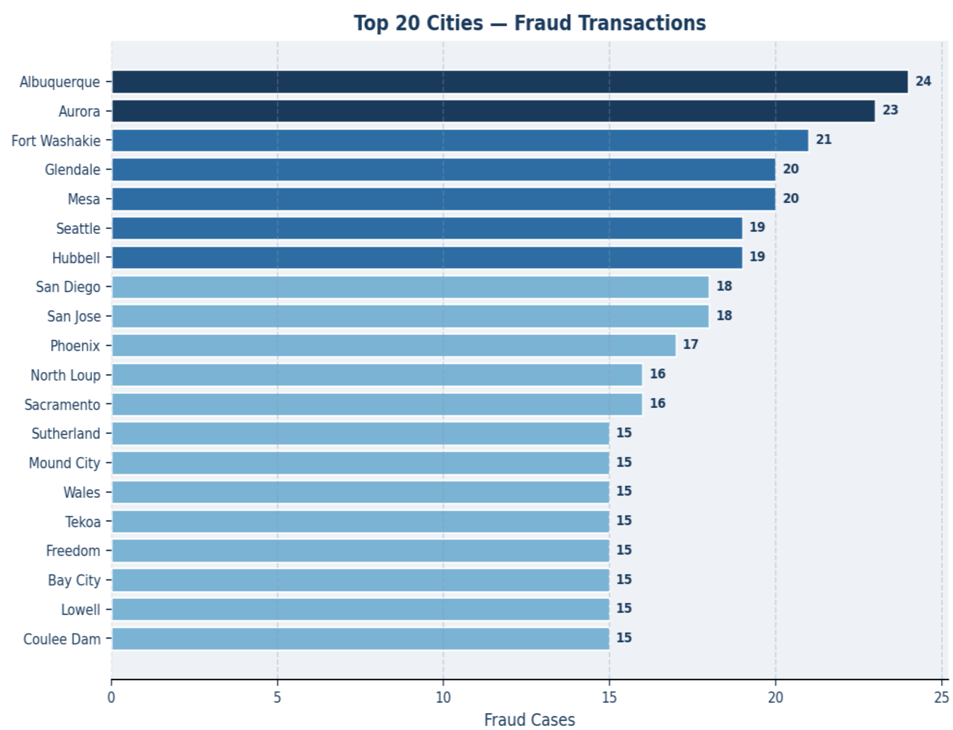

Figure 4 — Top 20 cities by fraud transaction count

Q5 — What does city population tell us about fraud?

Albuquerque (pop. 641,349) records the most fraud at 24 cases; Aurora (pop. 389,246) follows at 23. However, smaller cities like Altonah (pop. 302) also appear high in the fraud list, indicating that city population alone is not a definitive fraud predictor.

Q6 — What does the state-wise fraud analysis show?

California (CA) leads with 402 fraud cases out of 942 total transactions in that state. Missouri (MO) follows with 262, Nebraska (NE) 216, and Oregon (OR) 197. Hawaii (HI) records the fewest with only 16 cases. Fraud is highly concentrated in CA and MO.

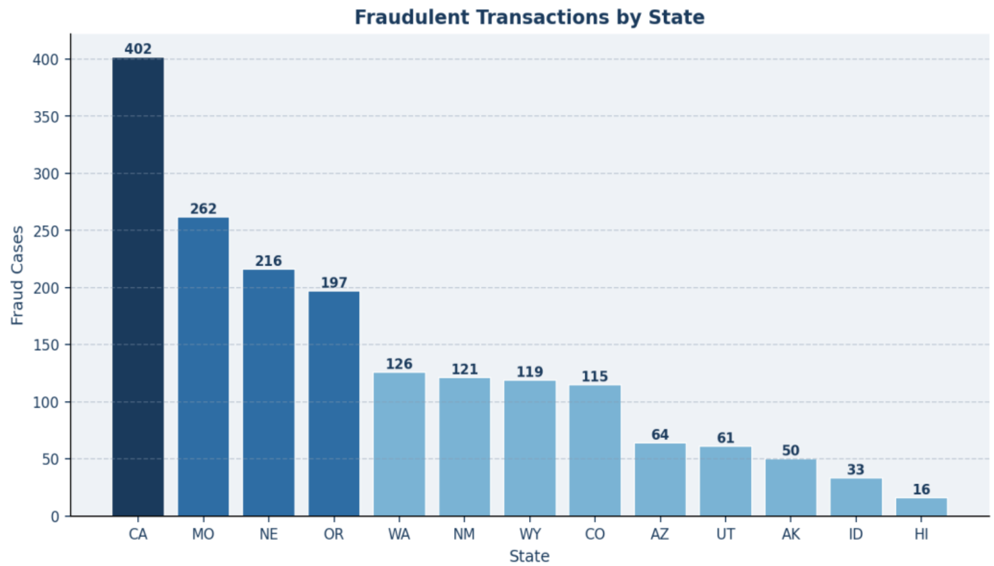

Figure 5 — Fraudulent transactions by state (CA dominates with 402 cases)

Q7 — What does the hourly fraud pattern show?

Fraud peaks sharply at 23:00 (459 cases) and 22:00 (452 cases). Early morning hours (00:00–03:00) are also elevated at 140–168 cases. Daytime hours (04:00–21:00) are significantly safer, with counts mostly below 25. Fraud clearly concentrates when monitoring is lowest.

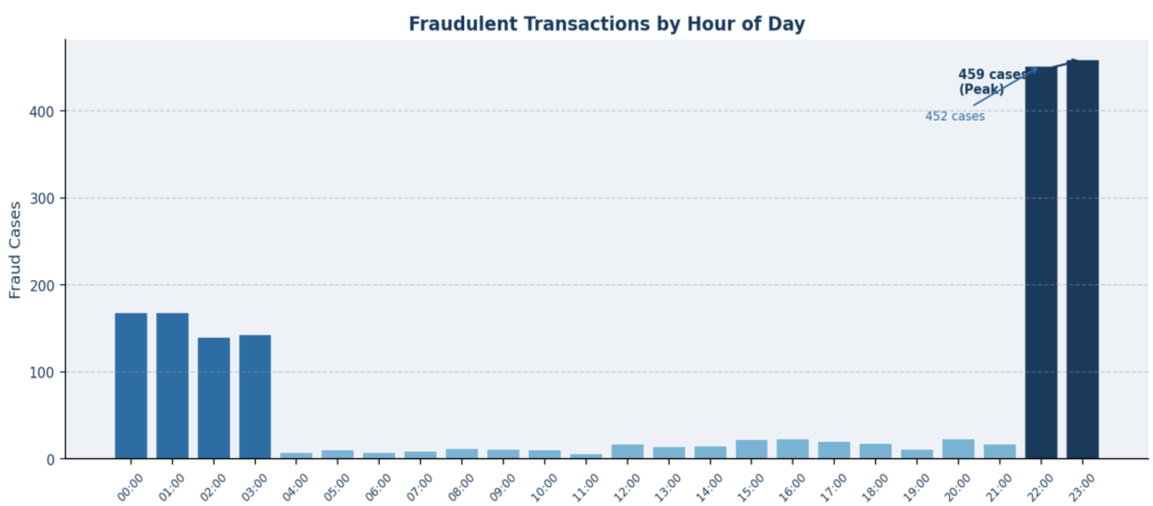

Figure 6 — Hourly fraud distribution showing sharp peaks at 22:00–23:00

Q8 — What does the monthly fraud pattern show?

March records the highest fraud (220 cases), followed by September (197) and February (169). June (92) and November (83) show the lowest activity. Fraud is unevenly distributed across months, peaking in spring and early autumn.

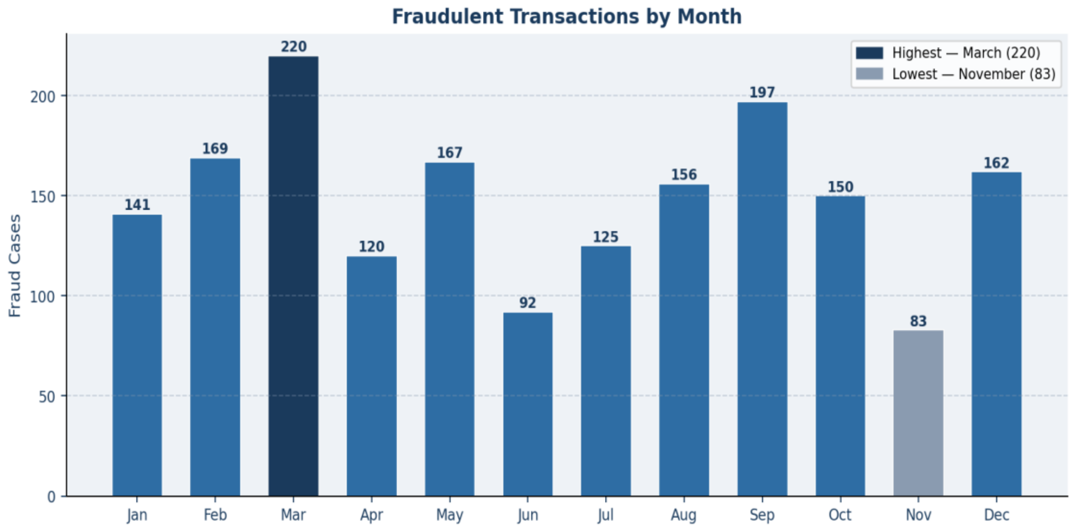

Figure 7 — Monthly fraud distribution (peak in March, lowest in November)

Q9 — What does the yearly fraud pattern show?

2019 recorded 978 fraud cases vs. 804 in 2020 — a 17.8% decline. This may reflect improved detection measures, changes in cardholder behavior during the COVID-19 pandemic, or reduced transaction volumes in 2020.

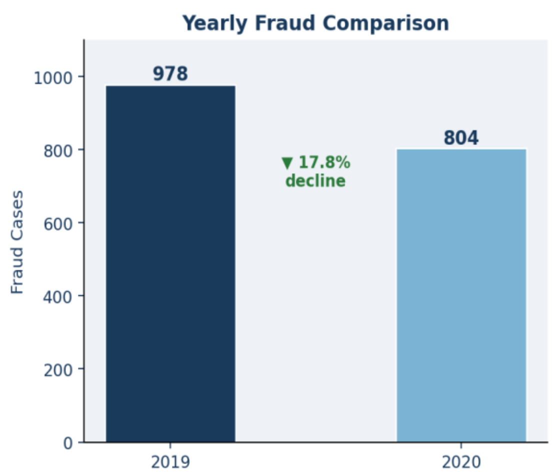

Figure 8 — Year-on-year fraud comparison (978 in 2019 vs 804 in 2020)

Q10 — What does the correlation between amount and fraud show?

The Pearson correlation between transaction amount (amt) and fraud (is_fraud) is 0.64, indicating a moderate positive relationship. As transaction amount increases, fraud probability tends to increase. However, this single feature is not sufficient for reliable detection — multiple features are required.

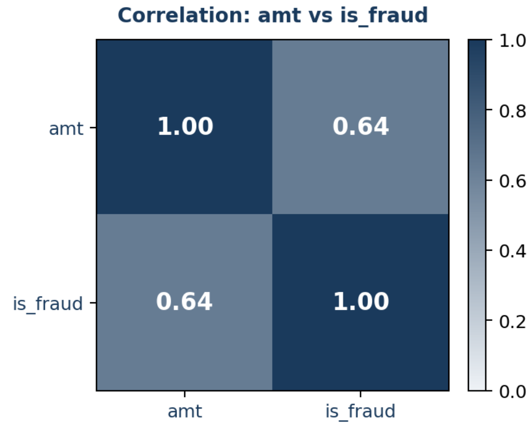

Figure 9 — Correlation heatmap: amt vs is_fraud (r = 0.64)

7. Machine Learning Model

7.1 Train-Test Split

The 4,000-row balanced dataset was split 80/20 — 3,200 training rows and 800 test rows — using random_state=42 for reproducibility.

7.2 One-Hot Encoding

The category and state columns contain text. A ColumnTransformer with OneHotEncoder (handle_unknown='ignore') converted them to numeric format, making them readable by the ML algorithm.

7.3 Random Forest Classifier

Random Forest was selected for its robustness to overfitting, ability to handle mixed feature types, and strong performance on binary classification tasks.

Parameter	Value	Purpose
n_estimators	300	Number of decision trees in the forest
max_depth	6	Limits tree depth to prevent overfitting
min_samples_split	5	Minimum samples required to split a node
random_state	42	Ensures reproducibility across runs

7.4 Pipeline

A scikit-learn Pipeline chained the ColumnTransformer (trf1) and RandomForestClassifier (trf2), ensuring consistent preprocessing during both training and inference.

8. Model Evaluation & Results

Metric	Score	Notes
Test Accuracy	87.50%	Evaluated on 800 held-out test samples
Cross-Validation Accuracy	88.09%	5-fold CV on training set — no overfitting detected
Top Feature	amt (r = 0.64)	Transaction amount — strongest predictor
Sample Prediction	Non-Fraudulent	Test: category=misc, amt=101, state=AZ, age=56

- Transaction amount (amt) has the strongest predictive impact — higher amounts correlate with fraud
- Engineered features (hours, distance, age, day) meaningfully boosted model performance beyond raw columns
- 5-fold cross-validation confirmed consistent, generalizable results with no significant overfitting
- The pipeline correctly classifies unseen transactions demonstrating real-world applicability

9. Conclusion

This project delivered a complete end-to-end fraud detection pipeline — from a severely imbalanced raw dataset to a validated, generalizable machine learning model achieving 87.50% accuracy.

- Undersampling was critical to preventing the model from ignoring the minority fraud class
- EDA revealed clear patterns: late-night hours (22–23), grocery and online categories, and states CA & MO are highest-risk
- Feature engineering (age, distance, hours, day) enriched the dataset significantly beyond raw columns
- Transaction amount is the strongest individual predictor (r = 0.64), but multiple features together are necessary
- Random Forest with cross-validation delivered reliable, generalizable fraud classification (88.09% CV accuracy)
- This model serves as a strong foundation for real-world fraud screening with scope for SMOTE, deep learning, or real-time streaming

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